ANSWERS

1.

Question	Answer	Marks
1(a)	<u>uniform / constant</u> acceleration (for 6 s) / acceleration of 3.7 <u>m / s^2</u>	**B1**
	after 6(.0 s) constant speed / zero acceleration	**B1**
1(b)	(a =) (v–u) / t in any form numerical or algebraic **or** 3.7 (m / s^2) seen	**C1**
	(m =) F / a **or** (m =) Ft / v in any form numerical or algebraic	**C1**
	490 (kg)	**A1**
1(c)	area under graph **or** average speed × time numerical or algebraic	**C1**
	66 (m)	**A1**

2.

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	**B1**
	deceleration • has (average) value 0.39 m / s^2 • lasts for 40 s • is non-uniform	**B1**
1(b)(i)	friction (with road) **and** <u>air</u> resistance **or** (air) drag	**B1**
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) **or** (graph shows) deceleration is not constant / decreases **or** non-uniform decrease in speed	**B1**

3.

Question	Answer	Marks
7(a)	acceleration / increase in speed **and** gravity / weight mentioned	**B1**
	air resistance / friction increases (with speed)	**B1**
	terminal / constant velocity reached **or** air resistance (becomes) equal to accelerating force	**B1**
7(b)	(distance travelled =) area under the graph **or** (x =) ½(u + v)t **or** ½ × 15.2 × 4.0	**C1**
	35 – (½ × 15.2 × 4.0) **or** 30.4 **or** 30	**C1**
	4.6 m	**A1**
7(c)(i)	(magnitude of deceleration =) gradient **or** Dv / Dt **or** 15.2 / 4.0 **or** equivalent numbers (from graph) in same ratio	**C1**
	15.2 / 4.0 **or** equivalent numbers in same ratio	**C1**
	3.8 m / s^2	**A1**
7(c)(ii)	(F =) ma **or** (46 + 9.0) × 3.8 **or** 55 × 3.8	**C1**
	210 N	**A1**
7(c)(iii)	<u>from</u> kinetic energy (and from gravitational potential energy)	**B1**
	<u>to</u> thermal energy **and** no intermediate energy	**B1**
7(d)	due to his mass **or** mass mentioned	**B1**
	(his) mass resists a change to its state of motion	**B1**

4.

Question	Answer	Marks
1(a)	(distance =) area (under graph) **or** 77×56 **or** $\frac{1}{2} \times 77 \times 70$ **or** 4300 **or** 2700 **or** $\frac{1}{2} \times 77 \times (56 + 126)$	**C1**
	$77 \times 56 + (\frac{1}{2} \times 77 \times 70)$ **or** 4300 + 2700 **or** $\frac{1}{2} \times 77 \times (56 + 126)$	**C1**
	7000 m **or** 7.0×10^3 m **or** 7.0 km	**A1**
1(b)(i)	(a =) $\Delta v / \Delta t$ **or** 77 / 70 (etc.)	**C1**
	1.1 m / s^2	**A1**
1(b)(ii)	(F =) ma **or** $3.8 \times 10^5 \times 1.1$	**C1**
	4.2×10^5 N	**A1**

5.

Question	Answer	Marks
7(a)(ii)	area under curve **or** formula for area of a triangle / trapezium **seen** **or** working shown used	**B1**
	(accelerating) 80 (m) **or** (at constant speed) 160 (m) **or** (decelerating) 32 (m) seen	**C1**
	(total distance) 270 **or** 272 m	**A1**
7(a)(iii)	(average speed) = <u>total</u> distance / (total) time algebraic or numerical	**C1**
	11 m / s	**A1**
7(a)(iv)	horizontal line drawn from 0 to 24 s at any level **or** horizontal line at the value calculated in **7(a)(iii)** for at least 16 s	**B1**
	horizontal line from 0 to 24 s at the value calculated in **7(a)(iii)**	**B1**

6.

Question	Answer	Marks
1(a)	scale vector diagram showing 12 N, 36 N and correct resultant force in correct direction	**B1**
	38 ± 1 kN	**B1**
	018 ± 3°	**B1**
1(b)(i)	(only in uniform acceleration) same <u>increase</u> / <u>change</u> in speed / velocity **or** has a constant acceleration **or** <u>speed-time graph</u> is a straight line	**B1**
	(only in uniform acceleration) same <u>increase</u> / <u>change</u> in speed / velocity in <u>same time</u> (period) **or** same rate of change of speed / velocity (with time) **or** <u>change in</u> speed / velocity is proportional to <u>time</u>	**B1**
1(b)(ii)	(a =) F / m in any way, algebraic or numerical	**C1**
	0.63–0.66 m / s^2	**A1**

7.

Question	Answer	Marks
2(a)	1.4 s	**B1**
2(b)(i)	same change in speed / velocity in the same time	**B1**
2(b)(ii)	slope / gradient changes	**B1**
2(c)	any clear statement or any use of area under graph **or** $d = s \times t$ **or** an (average) speed × time	**C1**
	calculation of any area between 1.4 and 7.0 s on Fig.2.1 e.g. 14 × 2.6; 36.4; 4 × 3; 12; ½ × 12 × 2.6; 15.6; 2.6 × 8; 20.8 (m) seen	**C1**
	48 m	**A1**

8.

Question	Answer	Marks
1(a)(i)	accelerating **or** increasing speed	**B1**
	decreasing acceleration **or** speed increasing at a decreasing rate **or** reaches a constant speed	**B1**
1(a)(ii)	<u>initially</u> acceleration is due to force of gravity / weight (only) **or** force of gravity / weight larger than air resistance from (t = 0 to ≈ 7 s)	**B1**
	air resistance increases (with speed)	**B1**
	resultant force decreases / becomes zero **or** forces balance **or** air resistance equals weight	**B1**
1(b)	two different pairs of co-ordinates from <u>straight</u>-<u>line</u> section of graph seen	**C1**
	correct use of co-ordinates in a division	**C1**
	48 – 52 m / s	**A1**

9.

Question	Answer	Marks
7(a)	(between t = 0 and t = 10 s) it is accelerating **or** speed / velocity increasing	**B1**
	between t = 35 s and t = 40 s, it is decelerating **or** speed / velocity decreasing	**B1**
	magnitude of acceleration less than magnitude of deceleration **or** between t = 0 and t = 10 s starts from rest / zero speed **or** between t = 35ms and t = 40 s it finishes at rest / with zero speed	**B1**
7(b)(i)	two pairs of co-ordinates from the straight-line section	**C1**
	appropriate division using candidate's coordinates	**C1**
	11.7 ⩽ speed ⩽ 12.0 m / s	**A1**
7(b)(ii)	<u>total</u> distance / <u>total</u> time **or** (distance =) 390 (m) **and** 39 (s) ⩽ (time =) ⩽ 40 (s)	**C1**
	9.7 to 10.0 m / s	**A1**

10.

Question	Answer	Marks
1(a)	constant gradient	**B1**
1(b)	use of area (under graph) **or** $\frac{1}{2}$vt **or** average speed 4.5 (m/s)	**C1**
	9(.0) m	**A1**
1(c)	line of decreasing slope after 2 s	**B1**
	horizontal line at 12 m/s from 5.5 to 8 s	**B1**
	line, curved or straight, from (8, 12) to (11,10)	**B1**

11.

Question	Answer	Marks
1(a)	weight is directly proportional to mass **or** $W = mg$	**B1**
	a larger force acts on the more massive body **or** $a = F / m$ **or** $a = W / m$	**B1**
	mass cancels **or** acceleration inversely proportional to mass	**B1**
1(b)(i)	(Δv =) at **or** 1.6×1.5	**C1**
	2.4 m / s	**A1**
1(b)(ii)	2.4 marked on y-axis **and** straight line from (0, 0) to (1.5, 2.4)	**B1**
1(b)(iii)	$\frac{1}{2} \times 2.4 \times 1.5$ **or** area under the graph mentioned / attempted	**C1**
	1.8 m	**A1**

12.

Question	Answer	Marks
1(a)(i)	58 km	**B1**
1(a)(ii)	0.8(0) hr **or** 48 min	**B1**
1(b)	steep(er) / larg(er) slope / gradient **or** travels a larger distance in the same / unit time **or** takes a shorter time to travel the same distance **or** covers more distance in less time	**B1**
1(c)	(s =) d / t numerical or algebraic; 58 / 4	**C1**
	15 km / hr	**A1**

13.

Question	Answer	Marks
9(a)(i)	10.2–10.4 (cm) **or** 50 (km) **or** 15 (km) seen	**C1**
	51–53 km	**A1**
	direction (0)72–74° **or** N 72–74 °E etc.	**B1**
9(a)(ii)	(distance) has no direction **or (**distance) is a scalar	**B1**
9(a)(iii)	changes direction **or** goes round a corner	**B1**
	velocity changes **or** velocity depends on direction **or** acceleration / force is towards the centre of circle / centripetal	**B1**

14.

(a) points plotted correctly at (1,0.5) (2,1.6) (3,3) (4,5) and (5,7) **B1**
smooth curve from origin **B1**

(b) (i) straight line **B1**
or gradient / slope constant
travels equal distances in same time **B1**
or gradient equals speed

(b) (ii) weight **and** force (upwards) from oil / liquid **B1**
forces balanced / cancel / are equal (and opposite) **B1**
or no resultant (force)

15.

(a) distance measured (between 2 marks) with metre rule or tape measure **B1**
time measured (between 2 marks) with stopwatch / stopclock / timer **B1**
(average speed =) (total) distance / time **B1**

(b) **(i)** decrease in speed
or deceleration **B1**

(b) **(ii)** (Δv =) a $\times$ t in any form numerical or algebraic **C1**
4.5 m / s **A1**

16.

Question	Answer	Marks
1(a)	(*F* =) *ma* **or** 35 × 2.6	**C1**
	91 N	**A1**
1(b)(i)	straight line from origin to dashed line / t_1	**B1**
	gradually decreasing gradient until the line is horizontal	**B1**
1(b)(ii)	area under the line **or** area between line and *x*-axis	**B1**

17.

(a) **(i)** (a=) v(–u) / t **or** 25 / 14 **C1**

1.8 m / s^2 **A1**

(ii) initial straight line from (0,0) to (14,25) **B1**
gradient of line decreases after 14 s
and flat from (70,55) to (80,55) **B1**

18.

(a) ($v = u +$)at **or** 3.4 × 5.0 C1
17 m / s A1

(b) **(i)** 0 **or** zero **or** no resultant force B1

(ii) straight line of positive gradient from (0, 0) B1
horizontal line at v > 0 **and** after initial acceleration B1
straight line from (0, 0) to (5.0, 17) **and**
straight line from (5.0, 17) to at least (15.0, 17) B1

(iii) calculate the area **under** the graph **or** area of trapezium B1 [7]

19.

(a) **(i)** (a =)$\Delta v / t$ **or** 95 / 0.011 C1
8.6(3636) × 10^3 m / s^2 A1

(ii) (F =)ma **or** 0.018 × 8.63 × 10^3 C1
150 / 155(.4545) / 160 N A1

(b) line from (0, 0) to (0.011, 95) with decreasing gradient B1
(becomes) horizontal at (0.011, 95) B1

20.

(a)	**(i)**	60 m	B1
	(ii)	12 s	B1
(b)	**(i)**	straight line from origin to 200 m at 40 s any line straight or curved from (40,200) to (60,500)	B1 B1
	(ii)	$s = d / t$ or 500 / 60 8.3 m / s	C1 A1

21.

(a)	**(i)**	distance (travelled) per second **or** speed distance (travelled) per second / speed in a given direction **or** displacement / time **or** change in displacement per unit time **or** displacement (travelled/covered) per unit time **or** rate of change of displacement		C1 A1
	(ii)	opposite direction		B1
	(iii)	**1**	value seen for v and corresponding value of t $0 < t \leqslant 1.4$ and $0 < v \leqslant 14$ (a=) v-u / t algebraic or numerical equation 10 m / s^2	C1 C1 A1
		2	sensible comment	A1

22.

(a)		changing speed / velocity change in speed / velocity / time constant **or** (v–u) / t constant **or** constant / equal rate of change of speed / velocity	C1 A1
(b)		(a vector quantity has) direction	B1
(c)	**(i)**	**1.** X between $t \geq 0$ and $t < 10$ s **2.** Y between $t > 20$ s and $t < 30$ s **3.** Z between $t > 10$ s and $t < 20$ s **or** between $t > 30$ s and $t < 40$ s	B1 B1 B1
	(ii)	**1.** **two** speed values from graph between 15 and 35 s (±1 mm) **two** corresponding time values from graph between 15 and 35 s (±1 mm) **or** (a =)$\Delta v / t$ 500 m / s^2	C1 C1 A1

23.

(a)	straight line from (0, 0) to (3, 2.4) horizontal line from 3 s to 8 s straight line from **end of a horizontal line** to zero in 1 s	[B1] [B1] [B1]
(b)	constant / same increase in velocity **or constant** change in velocity constant / same increase in velocity per sec / unit time	[C1] [A1]
(c)	occurs in a short(er) time **or** acceleration took 3 s and deceleration took 1 s	 [B1]
(d)	(d =) speed × time numerical or algebraic **or** area under graph clear 1.2 × 3 **or** 3.6 (m) **or** 2.4 × 5 **or** 12 seen 15.6 m **or** 16 m	[C1] [C1] [A1]
(e)	**(i)** mgh seen in any algebraic or numerical form, e.g. 30 × 10 × 1.6 480 J	[C1] [A1]
	(ii) heat or thermal energy or sound produced **or work done** against friction / air resistance	 [B1]
(f)	at least two distances and corresponding times mentioned	[C1]
	how the **actual measurement** is made, e.g. **(any one from)** • make mark on ground every second and **measure** distances • note video position every sec and use a scale to **find** distances • make mark on ground every meter and **measure / take** time as girl passes	 [A1]
	how constant speed is proved using measurement, e.g. **(any one from)** • same distance between each position for the same time interval • same time interval for equal distances • Δd / Δt constant **or** slope of distance-time graph constant	 [B1]

24.

(a)	**(i)** start at origin **and** not horizontal gradient (gradually) decreasing (**ignore** sudden decrease) (not if part of curve above horizontal section) final horizontal section (≥ 1 cm) (not if v is shown as ≠ 40 m/s)	B1 B1 B1
	(ii) area **under** the graph **or** count squares **under** graph between t = 0 and horizontal section **or** when speed is changing **or** calculate equivalent distance to 1 cm^2 (after counting squares)	M1 A1

25.

(a)	travels further in each second / in same time / between images	B1
(b)	(s=) d/t in any form algebraic or numerical	C1
	40 cm/s; 0.4(0) m/s	A1

26.

(a)		72 m/s	B1
(b)	**(i)**	area (under graph) **or** ½ base × height **or** ½vt **or** ½ × 9 × 72	C1
		320/324 m	A1
	(ii)	**change** in velocity/time **or** $\Delta v/t$ **or** 72/9	C1
		8(.0) m/s^2	A1

27.

(a)	**(i)**	curve with decreasing gradient from origin to 50 m/s at 10 s	B1
		constant speed from 10 to 20 s	B1
		decrease to 5 m/s at 25 s	B1
		constant speed from 25 s until at least 30 s	B1
	(ii)	gradient/slope not constant/decreases // graph curves // graph not a (straight) line // increase (in speed) per second/unit time not equal	B1
(b)		any mention of **air** resistance/drag/upward force	B1
		(initially) force upwards larger than force downwards // resultant force upwards	B1
		air resistance decreases (with fall in speed)	B1
		(at constant speed) air resistance/friction/drag equals weight // forces (up and down) balance // zero resultant force	B1
(c)		500 m	B1
(d)	**(i)**	(a =) $\frac{v-u}{t}$ in any numerical or algebraic form, e.g. 45/5	C1
		9(.0) m/s^2 ecf **(a)(i)**	A1
	(ii)	($F = ma$) in any numerical or algebraic form, e.g. 60 × 9 ecf **(i)**	C1
		540 N	A1
	(iii)	area **under** graph/line/curve	B1

28.

(a)	**(i)**	11.5 m/s	B1
	(ii)	decrease in speed // negative gradient	C1
		equal changes/decreases in speed in the same time // const. neg. grad. on v-t graph	A1
(b)	**(i)**	flat line at 18 m/s from t = 0 to 15	B1
		constant slope downwards parallel to initial line (by eye)	B1
	(ii)	greater area **under** graph // higher **initial/average** speed	B1

29.

(a) velocity has a direction/is a vector **or** speed does not have a direction/is not a vector **or** displacement/time **and** distance/time
(**ign** speed is a scalar) B1

(b) **(i)** (–) 47 m/s B1

(ii) (a =) v/t **or** 47/0.0013 C1
(–) 3.6(1538 etc.) × 10^4 m/s^2 A1

30.

(a) **(i)** any **one** time between 1.60 and 2.50 s **or** range of correct values B1

(ii) any **one** time between 0.75 and 1.65 s **or** range of correct values B1

(iii) 2.5(0) s B1

(b) area (under graph) **or** ½bh **or** ½gt^2 **or** ½ × 0.75 × (7.3 to 7.5) C1
2.7(375) to 2.8(125) m A1

31.

(a) (speed) increases or (paper) accelerates
(speed) becomes constant/uniform or acceleration zero (after 0.5 s) B1 B1

(b) any clear change in distance/time or 1.87 (m/s) (allow 1.9)
2.3–2.5 m/s C1

32.

(a) parachute opens or speed drops from (50 to 5 m/s) **or** decelerates (**e.g.** uniformly) **and** lands/hits ground **or** speed becomes 0 or stops (**e.g.** decelerates) B1

(b) accelerates **or** speed increases (**not** increasing acceleration) B1
acceleration decreases (to 0) **or** speed becomes constant B1

(c) forces balance/cancel **or** no resultant **or** equal and opposite (**not just** forces equal) B1
weight/gravity **and** air resistance/drag mentioned (**not** upthrust/friction) B1

(d) (d =) st **or** s=d/t or any speed x any time **or** area under graph C1
150 m A1

33.

(a) accelerates or speed increases **from rest/for 2-4s/for 8-20m** B1
then a constant/uniform speed **or** velocity B1

(b) between 7 and 8 m B1

(c) distance 80 (+ 2) or s= d/t in any algebraic or numerical form C1
7.3 **or** 7.27 **or** 7.273 m/s A1

34.

(a)	**(i)**	12 m/s	B1
	(ii)	16 s	B1
	(iii)	192 m **or** (i) × (ii)	B1
(b)		a = (v-u)/t in any format e.g. numerical (allow 4 clearly attributable wrong numbers) **or** gradient of v-t/the graph	C1
		2.7 (2/3 sig. fig. only, do not accept fraction, **cao**)	A1
		m/s^2	B1

35.

	any position before 50 m	B1
(b)	points plotted correctly at 4,8 and 12 s (± ½ square)	B1
	from origin to 4 sec **curve** drawn	B1
	from 4 to 12 sec straight line positive gradient	B1
	from 12 to 16 sec gradient decreases (but not –ve)	B1
(c)	speed/time	C1
	3 m/s	A1

36.

(a)	**speed** uniform or 20 m/s	**B1**
	stationary/not moving till 20 minutes or after 65 minutes **or** moves for 45 minutes	**B1**
	(**not** if inconsistent; all times ±2 min; **ignore** acceleration/deceleration periods)	
(b)	d = st any algebraic **or** area calculated	
	or 20 x 45, 20 x 90, 20 x 45 x 60, 20 x 90 x 60	**C1**
	54 000 m **or** 54 km	**A1**
(c)	any constant speed from **0 to 90** minutes (may stop at 90 or go down to axis)	**M1**
	10 m/s (no e.c.f. b)	**A1**

37.

(a)	distance traveled per unit time **or** in one second / distance ÷ time **or** rate of change of distance		B1
(b)	s = d/t in any algebraic or numerical form		C1
	any doubling of distance or final time		C1
	0.48 s	(allow 0.24s 2/3 accept 0.5s)	A1
(c)	60/0.48 (5)		C1
	123.75 accept 120, 123, 124	(ecf **(b)**)	A1